

Chapter 1 - Logic, Sets and Real Number System

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#Logic

Logic (derived from the word 'Logos' which means reason) is the science of reasoning. Logic tells us the truth and the falsity of the particular statement. Logic is the process by which we arrive at a conclusion from the given statement with a valid reason.

There are 2 types of reasoning:

1. Inductive reasoning (Reasoning a particular case from observation)
2. Deductive reasoning (Reasoning a particular case from a general case)

In logic, we use symbols for words, statements and their relations to get the required result.

Hence, logic is known as mathematical logic or symbolic logic.

#Statement

A statement is an assertion (declaration) expressed in words or symbols, which is either true or false but not both at the same time. In the context of logic, a statement cannot be imperative, interrogative or exclamatory.

Types of Statement	
On the basis of truth value	Open statement: An open statement is the statement whose truth or falsity can be decided after filling the gap in them or substituting the value of the variable.
	Closed statement: A closed statement is the statement whose truth or falsity cannot be decided. It does not have a gap as the open statement.
On the basis of structure	Simple statement: A simple statement is the statement which declares only one thing. It cannot be divided into two or more statements.
	Compound statement: A compound statement is the combination of two or more simple statements. It can be divided into two or more statements which are called component statements.

#Truth Value and Truth Table

Truth value refers to the truth or falsity of a statement. $T \rightarrow$ Statement is True and $F \rightarrow$ Statement is False

The truth value of a simple statement depends upon only the given statement. But in a compound statement, its truth value depends on both the component statement and the connectives used.

Truth table is the table presenting the truth values of the component statements together with their truth values.

#Logical Connectives

Logical connectives are the words or phrases which connect the component statements in a compound statement. All the logical connectives are given below:

Connective	Symbol	Definition	Truth Table		
Conjunction	$\wedge$	Two simple statements combined by the word 'and' (or an equivalent word), to form a compound statement, is known as conjunction of the given statements.	p	q	$p \wedge q$
			T	T	T
			T	F	F
			F	T	F
			F	F	F
Disjunction	$\vee$	Two simple statements combined by the word 'or' (or an equivalent word) to form a compound statement, is known as disjunction of the given statements.	p	q	$p \vee q$
			T	T	T
			T	F	T
			F	T	T
			F	F	F

Conditional (Implication)	$\Rightarrow$	Two simple statements combined by 'if...then' to form a compound statement, is known as conditional of the given statements.	p	q	$p \Rightarrow q$
			T	T	T
			T	F	F
			F	T	T
			F	F	T
Biconditional (Equivalence)	$\Leftrightarrow$	Two simple statements combined with 'if and only if' (abbreviated as iff) to form a compound statement, is known as biconditional of the given statements.	p	q	$p \Leftrightarrow q$
			T	T	T
			T	F	F
			F	T	F
			F	F	T

#Terminology

Negation	A statement which denies the given statement is known as the negation of the given statement. If p is a statement then $\sim p$ is said to be the negation of the statement p .
Tautology	A compound statement which is always true, whatever may be the truth values of its components, is known as tautology.
Contradiction	A compound statement which is always false, whatever may be the truth values of its components is known as contradiction.
Converse	If p and q are two simple statements then the conditional $q \Rightarrow p$ is said to be the converse of the conditional $p \Rightarrow q$.
Inverse	If p and q are two simple statements then the conditional $\sim p \Rightarrow \sim q$ is said to be the inverse of the conditional $p \Rightarrow q$.
Contrapositive	If p and q are two simple statements then the conditional $\sim q \Rightarrow \sim p$ is said to be the contrapositive of the conditional $p \Rightarrow q$.
Logically Equivalent	Two statements S_1 and S_2 are said to be logically equivalent if both have same truth values in the columns of the truth table. They are denoted by $S_1 \equiv S_2$.

#Laws of Logic

Primary Laws of Logic

Law of Excluded Middle	Only statement p or $\sim p$ is true. A statement cannot be both true and false at the same time.
Law of Tautology	The statement $p \vee \sim p$ is always true and hence a tautology. Thus, the disjunction of a statement and its negation is a tautology.
Law of Contradiction	The statement $p \wedge \sim p$ is always false and hence a contradiction. Thus, the conjunction of a statement and its negation is a contradiction.
Law of (Involution/Double Negation)	The negation of negation of a statement is logically equivalent to a given statement. Symbolically, $\sim(\sim p) \equiv p$
Law of Syllogism	If $p \Rightarrow q$ and $q \Rightarrow r$ then $p \Rightarrow r$. That is $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ This can be verified to be a tautology.
Law of Contraposition	$(p \Rightarrow q) \equiv ((\sim q) \Rightarrow (\sim p))$ The conditional and its contrapositive are logically equivalent.
Inverse Law	$(\sim p) \Rightarrow (\sim q) \equiv q \Rightarrow p$ The inverse and the converse of a conditional are logically equivalent.

Secondary Laws of Logic

Idempotent Law	$p \wedge p \equiv p$	$p \vee p \equiv p$
Commutative Law	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative Law	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$	$p \vee (q \vee r) \equiv (p \vee q) \vee r$
Distributive Law	$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
De-Morgan's Law	$\sim(p \wedge q) \equiv (\sim p \vee \sim q)$	$\sim(p \vee q) \equiv (\sim p \wedge \sim q)$

#Review of Set

Set and Its Notation

Set is the collection of well-defined objects. Each object of a set is called an element or member of the set.

Sets are usually denoted by capital letters such as A, B, X, Y, etc.

Membership of a Set

$\in$ → 'belongs to' or 'is an element' or 'is the member of'

$\notin$ → 'does not belong to' or 'is not the element of'

Special Types of Set

Empty/null/void set	A set having no element is called the empty set or null set or void set. It is denoted by the Greek letter ϕ (phi) or $\{\}$.
Finite set	A set containing a finite number of elements is known as a finite set.
Infinite set	A set which is not finite is known as an infinite set.

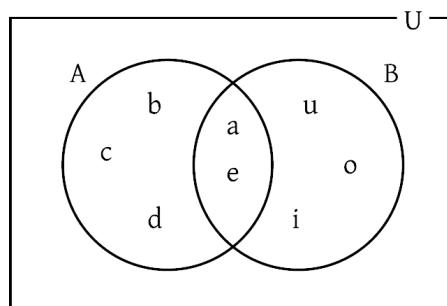
Relation between Sets

Subset and Superset	A set A is said to be a subset of the set B if every element of A is also an element of B . This relation is denoted by $A \subseteq B$. Here, B is called the superset of A and we write it as $B \supseteq A$. If every element of set A is also an element of set B but there is at least one element of B which is not an element of A , then A is known as the proper subset of B . This relation is denoted by $A \subset B$.
Equal sets	Two sets A and B are said to be equal or identical or same if they have the same elements. They are denoted by $A = B$. If $A \subseteq B$ and $B \subseteq A$ then $A = B$.
Intersecting sets	Two sets A and B are said to be intersecting if they have at least one element in common.
Disjoint sets	Two sets A and B are said to be disjoint if they have no elements in common.
Power set	The collection or the set of all possible subsets of a set is called its power set. Power set of a set A is denoted by $P(A)$ or 2^A .
Universal set	Universal set is the set which contains all the sets under discussion. It is denoted by U .

Venn Diagram

The diagrammatic representation of sets, set relations and set operations, is known as a venn diagram. It consists of a universal set U represented by a rectangle, subsets of U by the closed curves and the elements of the sets by the points within the closed curve.

E.g., The Venn-diagram of $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$ is present alongside.



Operations on Sets

Union	The union of two sets A and B is defined as the set of all elements which belong to A or B or both. Symbolically, $A \cup B = \{x: x \in A \text{ or } x \in B\}$ $x \in (A \cup B) \Rightarrow x \in A \text{ or } x \in B$ $x \notin (A \cup B) \Rightarrow x \notin A \text{ and } x \notin B$
Intersection	The intersection of two sets A and B is the set of all elements belonging to both sets A and B. Symbolically, $A \cap B = \{x: x \in A \text{ and } x \in B\}$ $x \in (A \cap B) \Rightarrow x \in A \text{ and } x \in B$ $x \notin (A \cap B) \Rightarrow x \notin A \text{ or } x \notin B$
Difference	The difference of two sets A and B is the set of all elements of A not belonging to B. Symbolically, $A - B = \{x: x \in A \text{ and } x \notin B\}$ $B - A = \{x: x \in B \text{ and } x \notin A\}$
Complement	The complement of a set A is the set of all elements in the universal set U that do not belong to A. Symbolically, $\bar{A} = \{x: x \in U \text{ and } x \notin A\}$ $x \in \bar{A} \Rightarrow x \notin A$ $x \in A \Rightarrow x \notin \bar{A}$
Symmetric Difference	The union of the differences $A - B$ and $B - A$ of two sets A and B is called the symmetric difference of A and B i.e. $A \Delta B = (A - B) \cup (B - A)$. Symbolically, $A \Delta B = \{x: x \in (A - B) \text{ or } x \in (B - A)\}$ $x \in (A \Delta B) \Rightarrow x \in A \text{ or } x \in B \text{ but } x \notin (A \cap B)$

#Theorems based on Set Operation

Properties of Inclusion and Equality Relations

1. $A \subseteq B, B \subseteq C \Rightarrow A \subseteq C$
2. $A = B \Rightarrow B = A$
3. $A = B, B = C \Rightarrow A = C$
4. $A \subseteq \emptyset \Rightarrow A = \emptyset$

Properties of Unions

1. $A \cup A = A$
2. $A \cup \emptyset = A$
3. $A \cup U = U$
4. $A \cup B = \emptyset \Rightarrow A = \emptyset \text{ and } B = \emptyset$
5. $A \cup B = B \cup A$
6. $(A \cup B) \cup C = A \cup (B \cup C)$

Properties of Intersections

1. $A \cap A = A$
2. $A \cap \emptyset = \emptyset$
3. $A \cap U = A$
4. $A \cap B = \emptyset \Rightarrow A = \emptyset \text{ and } B = \emptyset$
5. $A \cap B = B \cap A$
6. $(A \cap B) \cap C = A \cap (B \cap C)$

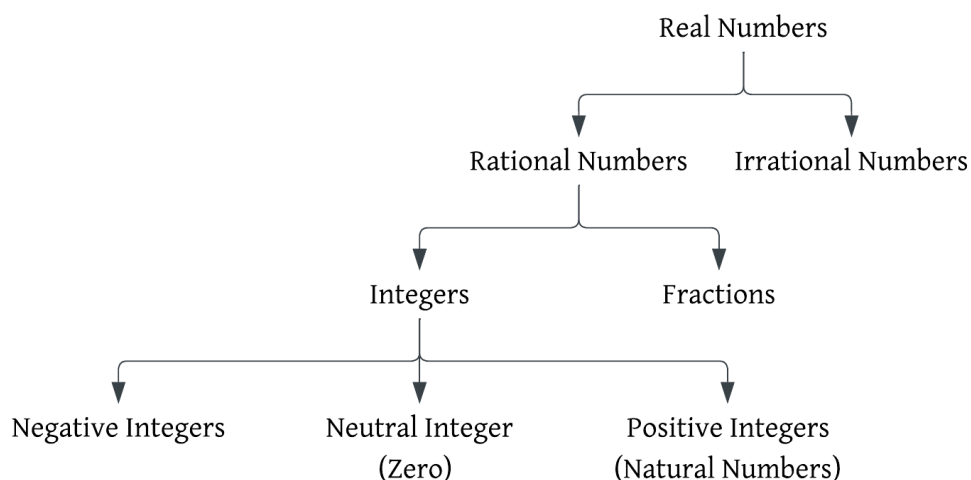
Miscellaneous Properties

1. Simple Distributive Laws
 - a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. Complement Laws
 - a. $A \cup \bar{A} = U$
 - b. $A \cap \bar{A} = \emptyset$
 - c. $\bar{\bar{A}} = A$
3. De-Morgan's Law
 - a. $\overline{A \cup B} = \bar{A} \cap \bar{B}$
 - b. $\overline{A \cap B} = \bar{A} \cup \bar{B}$
4. Special Distributive Laws
 - a. $A - (B \cap C) = (A - B) \cup (A - C)$
 - b. $A - (B \cup C) = (A - B) \cap (A - C)$
 - c. $A \cap (B - C) = (A \cap B) - (A \cap C)$

#Review of Real Numbers

Natural Numbers	The simplest and the most familiar unending chain of consecutive numbers are known as natural numbers. They are also known as counting numbers. The set of natural numbers is denoted by N.
Integers	The set of natural numbers together with their negatives including zero are known as Integers. The set of integers is denoted by Z. $Z^- \rightarrow$ Negative Integer $Z^0 \rightarrow$ Neutral Integer (Zero) $Z^+ \rightarrow$ Positive Integers
Rational Numbers	A number in the form of $\frac{p}{q}$ where p and q are integers and $q \neq 0$, is known as a rational number. The set of rational numbers is denoted by Q.
Irrational Numbers	Numbers which are not rational are known as irrational numbers. That is, a number which cannot be expressed in the form of $\frac{p}{q}$ where p and q are integers and $q \neq 0$, is known as an irrational number. The set of irrational numbers is denoted by $\bar{Q}$.
Real Numbers	The union of the set of rational and irrational numbers is known as real numbers. The set of real numbers is denoted by R.

Classification of Real Numbers



The family members of the real numbers have the following set relation:

$$N \subset Z \subset Q \subset R$$

#Properties of Real Numbers

1. Closure property:

$$\text{If } a, b \in \mathbb{R} \text{ then } \begin{cases} a + b \in \mathbb{R} \\ ab \in \mathbb{R} \end{cases}$$

2. Commutative property:

$$\text{If } a, b \in \mathbb{R} \text{ then } \begin{cases} a + b = b + a \\ ab = ba \end{cases}$$

3. Associative property:

$$\text{If } a, b, c \in \mathbb{R} \text{ then } \begin{cases} a + (b + c) = (a + b) + c \\ a(bc) = (ab)c \end{cases}$$

4. Identity Elements:

$$\text{For every } a \in \mathbb{R} \begin{cases} a + 0 = 0 + a = a \\ a \cdot 1 = 1 \cdot a = a \end{cases}$$

Hence, 0 and 1 are the additive and multiplicative identity of a .

5. Inverse Elements:

$$\text{For every } a \in \mathbb{R} \begin{cases} a + (-a) = (-a) + a = 0 \\ a \cdot a^{-1} = a^{-1} \cdot a = 1, (a \neq 0) \end{cases}$$

Hence, $(-a)$ and a^{-1} are the additive and multiplicative inverse of a .

#Order Property

The most basic order properties are:

- A real number a is said to be positive if $a > 0$ and negative if $a < 0$.
- If $a, b \in \mathbb{R}, a > 0, b > 0$ then $ab > 0$.

Besides the above property, we have the following properties as well:

1. The Axiom of Trichotomy (Trichotomy property):

If a and b are two real numbers then one and only of the following relations is holded

$$a < b, a = b, a > b$$

2. The Axiom of Transitivity (Transitivity property):

$$\text{If } a, b, c \in \mathbb{R} \text{ then } \begin{cases} a > b, b > c \implies a > c \\ a < b, b < c \implies a < c \end{cases}$$

3. Addition property:

$$\text{If } a, b, c \in \mathbb{R} \text{ then } \begin{cases} a > b \implies a + c > b + c \\ a < b \implies a + c < b + c \end{cases}$$

4. Multiplication property:

$$\text{If } a, b, c \in \mathbb{R} \text{ then } \begin{cases} a > b \xrightarrow{\times c, (c > 0)} ac > bc \\ a > b \xrightarrow{\times c, (c < 0)} ac < bc \\ a < b \xrightarrow{\times c, (c > 0)} ac < bc \\ a < b \xrightarrow{\times c, (c < 0)} ac > bc \end{cases}$$

#Representation of a Real Number in a Real Number Line

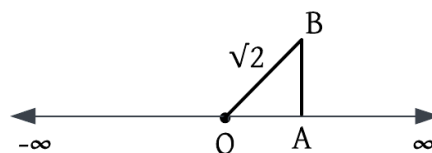
The straight line in which the real numbers are plotted by means of the points is called the real number line. Thus, any point on the real number line is said to be a real number. Since the real numbers are categorized into two types: Rational and Irrational numbers, broadly we have the following representations of real number:

Representation of a rational number in a real number line

1. Draw a horizontal straight line and mark a point as 0 (origin).
2. Mark 1 unit to the right of 0.
3. If the rational number is $\frac{p}{q}$ then divide the unit interval into q equal parts.
4. From 0, move p parts to the right (if positive) or left (if negative).
5. The obtained point represents the given rational number.

Representation of an irrational number in a real number line

1. Draw a number line and take $OA = 1$ unit.
2. At point A, draw a perpendicular $AB = 1$ unit.
3. Join OB. By Pythagoras' theorem: $OB = \sqrt{2}$
4. Point B represents $\sqrt{2}$ on the real number line.
5. Repeat step(2) at the new point obtained in step(4) for $\sqrt{3}, \sqrt{4}, \sqrt{5}$, etc.



Inequality: An inequality is a relation comparing two expressions that are not strictly equal, using symbols like $<$ (less than), $>$ (greater than), $\leq$ (less than or equal to), or $\geq$ (greater than or equal to). They define a range of possible values rather than a single solution.

#Interval

The set of points on the real number line between a and b , is known as an interval. An interval is denoted by I . An interval may or may not include the end points. On this basis, there are 4 different types of intervals:

Open interval	An interval not containing both the end points a and b , is known as an open interval. Symbolically, $(a, b) = \{x: a < x < b\}$
Closed interval	An interval containing both the end points a and b , is known as a closed interval. Symbolically, $[a, b] = \{x: a \leq x \leq b\}$
Left open interval	An interval not containing the end point a but containing the end point b , is known as the left open interval. Symbolically, $(a, b] = \{x: a < x \leq b\}$
Right open interval	An interval containing the end point a and not containing the end point b , is known as the right open interval. Symbolically, $[a, b) = \{x: a \leq x < b\}$

#Absolute Value

The absolute value (or modulus) of a number x , is a non-negative real number defined by

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Geometrically, $|x|$ is the distance from the origin to the number x .

Moreover, the distance between any two numbers a and b on the real number line is $|a - b| = |b - a|$.

Properties of (Absolute Values/Moduli)

1. Unary Property

- a. $|x| \geq 0$
- b. $|x| \geq x$ and $|x| \geq -x$
- c. $-|x| \leq x \leq |x|$

2. Binary Property

- a. $|x + y| \leq |x| + |y|$
- b. $|x - y| \geq |x| - |y|$
- c. $|xy| = |x||y|$
- d. $\left| \frac{x}{y} \right| = \frac{|x|}{|y|}$

3. Omission Property

- a. If $x \in R$ and a is any positive real number then $|x| \leq a \Leftrightarrow -a \leq x \leq a$